

Activity: Filter a SQL query

Scenario

In this scenario, you need to get specific information about employees, their machines, and the departments they're in. Your team needs this data to perform various tasks, such as running updates, posting a privacy notice in certain departments, and sending an alert to an employee with an issue on a machine.

You are responsible for finding the required information by querying a database. You'll add filters to your queries to locate the information more quickly.

Orientation: Know Your Data

Before you start querying, it is helpful to know exactly what the tables and columns look like. Run these two commands in your terminal to see the structure of the database

DESCRIBE machines;

DESCRIBE employees;

```
clear
MariaDB [organization]> DESCRIBE machines;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| device_id      | varchar(16)   | NO   | PRI | NULL    |       |
| operating_system | varchar(16)   | NO   |     | NULL    |       |
| email_client    | varchar(16)   | NO   |     | NULL    |       |
| OS_patch_date   | date          | NO   |     | NULL    |       |
| employee_id     | int(11)       | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.015 sec)

MariaDB [organization]> DESCRIBE employees;
+-----+-----+-----+-----+-----+-----+
| Field          | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| employee_id     | int(11)       | NO   | PRI | NULL    |       |
| device_id       | varchar(16)   | YES  |     | NULL    |       |
| username        | varchar(16)   | NO   |     | NULL    |       |
| department      | varchar(32)   | NO   |     | NULL    |       |
| office          | varchar(32)   | NO   |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
5 rows in set (0.001 sec)
```

What does `DESCRIBE` do? Think of `DESCRIBE` as a Table of Contents. It tells you exactly how a table is built without showing you all the rows of data. When you run it, you will see:

- **Field:** The exact name of the columns (e.g., `device_id`). You must use these exact names in your `SELECT` statements.
- **Type:** The kind of data allowed in that column (e.g., `int` for whole numbers or `varchar` for text).

You can run the `DESCRIBE` command at any time to refresh your memory on the column names for the tables.

Task 1. List all organization machines

In this task, you need to get a list of all organization machines and their operating systems. The data is contained in the `machines` table. You'll need to use the `SELECT` keyword to return specific columns.

- Run a SQL query to retrieve only the `device_id` and `operating_system` columns from the `machines` table.

```
MariaDB [organization]> SELECT device_id, operating_system FROM machines;
```

device_id	operating_system
a184b775c707	OS 1
a192b174c940	OS 2
a305b818c708	OS 3
a317b635c465	OS 1
a320b137c219	OS 2
a398b471c573	OS 3
a667b270c984	OS 1
a821b452c176	OS 2

How many rows were returned from the `machines` table? (You can view the number of rows at the bottom of the output.)

☐ 300



☒ 200

☐ 100

☐ 250

Submit

```
| z654a154b259 | OS 2 |
| z803a233b718 | OS 1 |
| z821a946b264 | OS 3 |
| z942a966b589 | OS 3 |
+-----+-----+
200 rows in set (0.198 sec)
```

Task 2. Retrieve a list of the machines with OS 2

In this task, you need to obtain a list of all machines with the 'OS 2' operating system because these machines need an update. To get this information, you'll run your first SQL query with a filter.

- Select all the records from the `machines` table with a value of 'OS 2' in the `operating_system` column. Replace the value X with the correct string:

```
MariaDB [organization]> SELECT device_id, operating_system
->
-> FROM machines
->
-> WHERE operating_system = 'OS 2';
```

device_id	operating_system
a192b174c940	OS 2
a320b137c219	OS 2
a821b452c176	OS 2
b157c491d493	OS 2

How many machines in the database use the OS 2 operating system?

☐ 200

☐ 88

☐ 44

☒ 80

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y347z204a710	OS 2
y765z123a548	OS 2
y976z753a267	OS 2
z451a308b518	OS 2
z654a154b259	OS 2

80 rows in set (0.142 sec)

Task 3. List employees in specific departments

In this task, you need to retrieve a list of all the employees in the Finance and Sales departments to obtain their office numbers. A notice about handling confidential financial information will be posted to these offices.

1. Filter the rows returned from department column in the employees table to include only employees from the 'Finance' department. Replace X with the appropriate column name and Y with the appropriate value to complete the filter:

```
MariaDB [organization]> SELECT * FROM employees WHERE department = 'Finance';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1007	h174i497j413	wjaffrey	Finance	North-406
1008	i858j583k571	abernard	Finance	South-170

What is the employee_id of the first row returned?

☐ 1049

☒ 1003

☐ 1001

☐ 1119

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2. Modify the previous query so that it returns employees who are in the 'Sales' department.

```
MariaDB [organization]> SELECT * FROM employees WHERE department = 'Sales';
```

employee_id	device_id	username	department	office
1009	NULL	lrodriqu	Sales	South-134
1011	1748m120n401	drosas	Sales	South-292
1024	y976z753a267	iuduik	Sales	South-215
1025	z381a365b233	jhill	Sales	North-115

Task 4. Identify employee machines

Your team recently discovered that there are issues with machines in the South building. In this task, you need to obtain certain employee and computer information.

A machine in 'South-109' has an issue. You need to determine which employee uses that computer so you can send them an alert.

1. Write a query to identify which employee uses the office in 'South-109'. (The data must be returned from the office column in the employees table.)

```
MariaDB [organization]> SELECT * FROM employees WHERE office LIKE 'South-109';
```

employee_id	device_id	username	department	office
1010	k2421212m542	jlansky	Finance	South-109

1 row in set (0.043 sec)



Which of the following employees uses the computer with the issue?

☐ tsnow

☐ jhill

☒ jlansky

☐ nmitchell

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2. Modify the query you used in the previous step so that it returns information on all the employees in the 'South' building. Use the `LIKE` operator with `%` in this query. For this task, you want to find all offices that **start** with the word 'South'. Your filter should look like this: `WHERE office LIKE 'South%'`;

```
MariaDB [organization]> SELECT * FROM employees WHERE office LIKE 'South%';
```

employee_id	device_id	username	department	office
1003	d394e816f943	sgilmore	Finance	South-153
1004	e218f877g788	eraab	Human Resources	South-127
1005	f551g340h864	gesparza	Human Resources	South-366

Which department does the first employee listed in the South building belong to?

- ☐ Sales
- ☐ Marketing
- ☐ Information Technology

☒ Finance

Submit